

The Paleo Diet and Human Paleodiets



The Paleo Diet® and Human Paleo Diets

Note: Make sure you have read both required sections of Gremillion and browsed the Paleo Diet® website prior to completing this section.

The Paleo Diet®

The Paleo Diet® is built on several core assumptions about what foods were eaten by humans during the period in which humans evolved: the Paleolithic (hence the “Paleo” diet). The creators of the diet summarize it as follows:

“The basic argument for following this way of eating is that, for most of human history, it is how we ate, and it subsequently shaped our genetics.

This is particularly true in terms of how we, as humans, respond to the foods we eat, to the prevention of disease, and the vitality of our species. While there have been massive changes in the food supply—starting with the agricultural revolution 10,000 years ago, then, the advent of dairy farming 5,000 years ago, and the more recent industrial revolution—our genetic makeup has not kept pace.

Humans do adapt to changes. For example, the genetic mutation that led to the development of adult lactase persistence (ALP)—allowing some of us to digest lactose in milk into adulthood—took place very recently, from an evolutionary standpoint. But when looked at in terms of human generations, we actually adapt very slowly. This can result in an incongruence between

our physiology and our environment. That is the case with the typical modern diet in the Western world. Over 70 percent of the foods we eat were introduced since our last major evolutionary change and we simply haven't caught up."

Source: "The Paleo Diet" website

(<https://thepaleodiet.com/try-the-paleo-diet/what-is-the-paleo-diet/>)

What is in the diet, and what is out?

"There are actually two simple ways to look at *what is Paleo*? The first is to focus on emulating the foods that our Paleolithic ancestors ate. The second, which is even simpler, is to look at what they did not eat and remove those foods from our diet. Let's briefly explore both.

There is some understandable confusion and debate around the concept of eating the foods we evolved around. The fact is there isn't a perfect record of the foods that were eaten. Furthermore, many of the foods that existed during Paleolithic times no longer exist. Fruits were wilder, smaller, and more nutrient dense. Hunted animals were often larger, and their body fat composition was different. And even if some of the foods do still exist, many of us don't live on the same continent as our ancestors. So, we don't have access to them.

This is why we sometimes refer to it as the *Paleolithic Diet Template*. The idea is to emulate our ancestral diet. Exactly what that looked like is still being debated; that's the debate we should be having. Nevertheless, there's no argument that at

its simplest, the Paleolithic diet consisted of a wide variety of vegetables, fruits, lean meats, fish, eggs, nuts, and seeds.”

Source: "The Paleo Diet" website

(<https://thepaleodiet.com/try-the-paleo-diet/is-it-paleo/>)

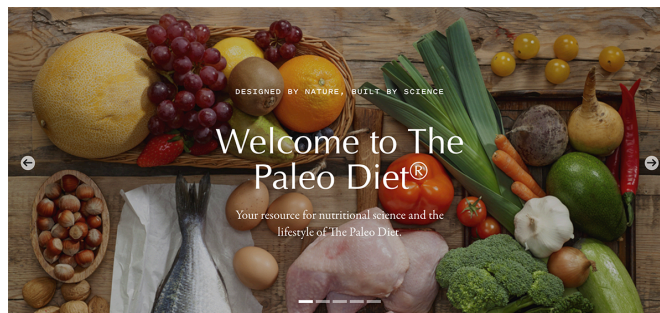
To summarize...

In: meats, fish, eggs, fruit, vegetables, nuts

Out: milk and milk products (cheese, yogurt, etc.); grains (including flours, beans, quinoa, wheat, corn, etc.); sugar; salt; “legumes and nightshades” (i.e., beans, peas, eggplants, tomatillos, potatoes)

Test Yourself

Look at this image, taken from the header image on the Paleo Diet® website, which shows only Paleo Diet®-compatible foods. Which of these foods were actually available in the Paleolithic? (Check all that are true.)



☒ A. **Incorrect:** Hazelnuts

☒ B. **Incorrect:** Melon (cantaloupe)

☒ C. **Correct:** Grapes Correct answer

☒ D. **Correct:** Kiwifruit Correct answer

☒ E. **Incorrect:** Orange

☒ F. **Correct:** Fish Correct answer

☒ G. **Correct:** Eggs Correct answer

☒ H. **Correct:** Chicken Correct answer

☒ I. **Incorrect:** Tomatoes

☒ J. **Incorrect:** Leeks

☒ K. **Incorrect:** Broccoli

☒ L. **Incorrect:** Avocado

☒ M. **Incorrect:** Chili Pepper

☒ N. **Incorrect:** Zucchini

☒ O. **Correct:** Garlic Correct answer

☒ This is false.

Hazelnuts — native to Europe and western Asia, hazel trees are widely cultivated but wild hazels produce similar nuts.

Melon (cantaloupe) — wild melons are native to South Asia but their fruits are much smaller and generally not sweet, though likely edible. The sugar content of modern melons was a trait introduced through generations of selective breeding.

Grapes — wild grapes are edible and sweet, though with numerous large seeds, and native to southwestern Asia.

Kiwifruit — while the large, fuzzy kiwifruits we know today are cultivated varieties, many wild species of kiwi (all native to East Asia) are edible, though much smaller.

Orange — sweet oranges are **hybrids**, the result of humans cross-breeding small, sweet mandarin oranges with large, sour pomelo. Even these edible wild ancestors would have only been available in small areas of Southeast Asia during the Paleolithic.

Fish — fish, even farmed ones, are wild animals; likely fish were more abundant and larger during the Paleolithic, before widespread overhunting by modern humans!

Eggs — chicken eggs are a product of modern breeding, but jungle fowl do lay clutches of several eggs of similar size (about 4.5 cm $\approx$ 2 inches in length).

Chicken — chickens are domesticated red jungle fowl, native to Southeast Asia. While chickens were not present in the Paleolithic, other birds of similar size (ducks, geese, quails, grouses, and fowl) could have been hunted.

Tomatoes — native to South America (where humans were not present earlier than 23,000 years ago, at the earliest), the wild ancestor of tomatoes is sweet and harmless—and the size of a pea! Learn more about the “pimp tomato” from the Smithsonian Magazine (<https://www.smithsonianmag.com/travel/why-wild-tiny-pimp-tomato-so-important-180955911/>).

Leeks — while wild leeks (native to western Eurasia) are edible, they are pencil-thin at their thickest and dark green. Fat, white

leeks are a product of centuries of selective breeding and specific cultivation methods.

Broccoli — a cultivated variety that is very different from its Paleolithic ancestor. Learn more in Dr. Warinner's talk below.

Avocado — native to Central America (where humans were not present earlier than 23,000 years ago, at the earliest). Wild avocados are similar to the variety we know today, but with a much thinner flesh in relation to the size of the pit.

Chili Pepper – native to South America (where humans were not present earlier than 23,000 years ago, at the earliest); wild peppers are the size of small grapes.

Zucchini — native to North and Central America (where humans were not present earlier than 23,000 years ago, at the earliest); the ancestor is a hard, striped gourd with bitter, inedible flesh. Likely only the seeds of this original plant were edible.

Garlic — native to western Asia, wild garlic species usually are edible, though pungent, with edible bulbs (what we eat as garlic), leaves (think chives), and stems (garlic scapes).

Hybrids are the result of two species being bred together, either naturally or through controlled breeding by humans. The names of hybrids insert an "×" between the genus and species names (e.g., *Citrus × sinensis*, the sweet orange) to denote that they are hybrids. Many hybrids are not fertile (e.g., a mule, a hybrid between a horse and a donkey, is infertile), but some are fertile when mating with their own hybrid species or other closely related species (as are the citrus fruits, which is

species (as are the other ones, which is why we have so many types).

Human Paleolithic Diets

The Paleo Diet® is built on several core assumptions about what foods were eaten by humans during the period in which humans evolved. But are these assumptions valid? Watch the 22-minute TEDx video below to see Prof. Christina Warinner, now of Harvard University, describe what evidence we have regarding the diets of humans during the Paleolithic, and why the Paleo Diet® may not accurately represent the food resources available during that time.



TEDx Talks

